# Zhen (Zoe) Qin


CONTACT INFORMATION

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## EDUCATION

**The Ohio State University** 08/2023 – 05/2028 (Expected)
*Ph.D. Candidate in Electrical and Computer Engineering*
- Advisors: Prof. Jia (Kevin) Liu and Prof. Yingbin Liang

**University of Southern California** 01/2020 – 12/2021
*M.A. in Applied Mathematics*

**Lanzhou University** 09/2014 – 07/2018
*B.S. in Mathematics and Applied Mathematics*

## RESEARCH INTERESTS

- **Optimization for Large-Scale ML:** Bilevel, multi-objective, distributed, and Muon-style optimization, with applications to LLM alignment and meta-learning.
- **LLM Agents and Agentic RL:** Stable and efficient reinforcement learning for multi-turn LLM agents.
- **AI Safety:** Preserving LLM safety through continued fine-tuning and model updates.

## SELECTED RESEARCH PROJECTS

**Efficient Optimization for Large-Scale Training** 2023 – Present

- Developed **BiMuon**, a Muon-style stochastic bilevel optimizer with convergence guarantees.
  - **Key idea:** Orthogonalizes hypergradient momentum to address spectral imbalance induced by lower-level curvature, with block-specific learning rates balancing upper-level progress and lower-level sensitivity, in exact SVD and Newton–Schulz variants.
  - **Results:** *CIFAR-100 (DeiT-S, 7.1M upper-level parameters):* the Newton–Schulz variant reached a 44% lower final validation loss than stocBiO+Adam (0.341 vs. 0.604 after 1,000 iterations) and hit a target loss of 0.75 1.9× faster in wall-clock time.
- Designed **DisMUSE**, a distributed Muon-style optimizer with convergence guarantees.
  - **Key idea:** Combines variance reduction with spectral updates to address noncommuting aggregation and orthogonalization, with exact SVD, Newton–Schulz, and low-rank variants trading off computation, communication, and accuracy.
  - **Results:** *CIFAR-10:* the Newton–Schulz variant improved test accuracy over FedAvg (89.92% vs. 84.21%) and trained 2.02× faster than the exact SVD variant; the low-rank variant (rank 32) cut simulated communication by 40.5% at an accuracy cost (89.69% → 85.08%). *Tiny Shakespeare:* the Newton–Schulz variant reduced validation perplexity from 12.25 (FedAvg) to 4.32.
- Proposed **DUET** (**ICLR 2025**), a decentralized bilevel algorithm for collaborative learning without lower-level strong convexity.
  - **Key idea:** Uses diminishing quadratic regularization for unique regularized lower-level solutions and gradient tracking to handle data heterogeneity.
  - **Results:** Established a finite-time convergence guarantee and evaluated the method on decentralized meta-learning (IID and non-IID data) and hyperparameter optimization.

**Multi-Objective Bilevel Learning for LLM Alignment** 2025 – 2026

- Co-authored an **ICML 2026** paper reformulating multi-task bilevel learning as equality-constrained multi-objective optimization to explore trade-offs among conflicting objectives.
  - **My role:** Designed and implemented the experimental evaluation.
  - **Key idea:** A weighted-Chebyshev penalty method explores objective trade-offs by varying preference weights, with finite-time guarantees for Pareto stationarity.
  - **Results:** Optimized HelpSteer data-mixing weights for LoRA fine-tuning across five criteria, achieving over 1000× the hypervolume of MoCo and MOML on Llama-3.2-1B-Instruct. Extended experiments to Llama-3.2-3B and Llama-3.1-8B with PyTorch DDP on 4×H100s.

**Other Optimization Research** 2022 – Present

- **SignRVR:** Developed epoch-anchored variance reduction for signSGD with random reshuffling to control gradient sign errors, improving convergence over SignRR and signSGD with test accuracy comparable to Adam.
- **DiSCO:** Integrates proximal consensus updates, diminishing regularization, and variance-reduced gradient tracking for constrained stochastic decentralized bilevel optimization without lower-level strong convexity.
- **Zeroth-order gossip:** Proved that the latest seed–scalar updates alone cannot exactly reconstruct decentralized gossip states, which depend on historical mixing.

**Ongoing Work**

- **Agentic RL:** Reproducing multi-turn agentic RL ($T^2$PO / GiGPO) for a Qwen3-4B web-shopping agent with verl and vLLM.
- **AI Safety:** Developing an evaluation pipeline with Qwen2.5-0.5B to test whether LLM backdoor repairs persist through subsequent fine-tuning.

## Open-Source Project

**Muon Paper Agent [GitHub]**

- Built an open-source agent that retrieves and ranks arXiv papers, generates LLM summaries, and delivers daily Slack digests via GitHub Actions, with configurable topics, keywords, and prompts.

## Publications

- **Zhen Qin**, Zhuqing Liu, Songtao Lu, Yingbin Liang, Jia Liu. "DUET: Decentralized Bilevel Optimization without Lower-Level Strong Convexity." ***ICLR***, 2025. [PDF]
- Zhiyao Zhang, Myeung Suk Oh, **Zhen Qin**, Jiaxiang Li, Xin Zhang, Jia Liu. "A Tale of Two Problems: Multi-Task Bilevel Learning Meets Equality Constrained Multi-Objective Optimization." ***ICML***, 2026. [arXiv]
- **Zhen Qin**, Zhiyao Zhang, Yingbin Liang, Jia Liu. "BiMuon: A Geometry-Aware Stochastic Bilevel Optimization Approach." *Under review*. [arXiv] [Code]
- **Zhen Qin**, Zhiyao Zhang, Yingbin Liang, Jia Liu. "DisMUSE: An Efficient Distributed Singular-Spectrum-Based Optimization Approach with Theoretical Convergence Guarantee." *Under review*. [arXiv]
- **Zhen Qin**$^*$, Zhishuai Liu$^*$, Pan Xu. "Convergence of Sign-based Random Reshuffling Algorithms for Nonconvex Optimization." *Under review*. ($^*$Equal contribution.)
- **Zhen Qin**, Zhuqing Liu, Yingbin Liang, Jia Liu. "Constrained Decentralized Bilevel Optimization over Networks: Beyond Lower-Level Strong Convexity." *Manuscript*.
- **Zhen Qin**, Zhe Li, Bicheng Ying, Haibo Yang, Yingbin Liang, Jia Liu. "Why Dimension-Free Zeroth-Order Communication Does Not Directly Extend to Decentralized Gossip." *Manuscript*.

## Work Experience

**Teaching Assistant, The Ohio State University** Fall 2026
ECE 8101: Nonconvex Optimization for Machine Learning (Prof. Jia Liu)

**Graduate Research Associate, The Ohio State University** 08/2023 – Present
Optimization for large-scale ML and LLM alignment (ICLR 2025, ICML 2026).

**Research Intern, Duke University (Prof. Pan Xu)** 01/2022 – 05/2023
Analyzed signSGD with random reshuffling and proposed SignRVR (co-first-author paper under review).

## Programming and Skills

**Languages:** Python, LaTeX.
**ML / LLM:** PyTorch (DDP, FSDP), Hugging Face Transformers, LoRA, vLLM, verl, Ray.
**Tools:** Git, Linux, multi-GPU (H100) servers, GitHub Actions, OpenAI API.

## Academic Services

**ICML Gold Reviewer, 2026**
**Conference reviewer:** AISTATS 2024–2027; ICLR 2025, 2027; ICML 2026.
**Journal reviewer:** IEEE Trans. Cognitive Communications and Networking; IEEE/ACM Trans. Networking; IEEE Trans. Information Theory.